Problem E2. Tungsten Filament *(13 points)*

Part A. Filament diameter *(1.5 points)*

Sketch the measurement setup:

For filament diameter d, directly measured quantities and calculations:

Filament diameter d and its uncertainty:

Part B. Filament's resistance *(2 points)*

Draw measurement circuit(s):

For filament room temperature resistance R, directly measured quantities and calculations:

Filament resistance R and its uncertainty

Filament length l and its uncertainty:

Part C. Current–voltage curve *(2.5 points)*

Draw measurement circuit(s):

Measurements (you don't have to fill the entire table):

Filament temperature before it broke T:

STUDENT No

ANSWER SHEET

PROBLEM E2

GPhO Riyadh 2016

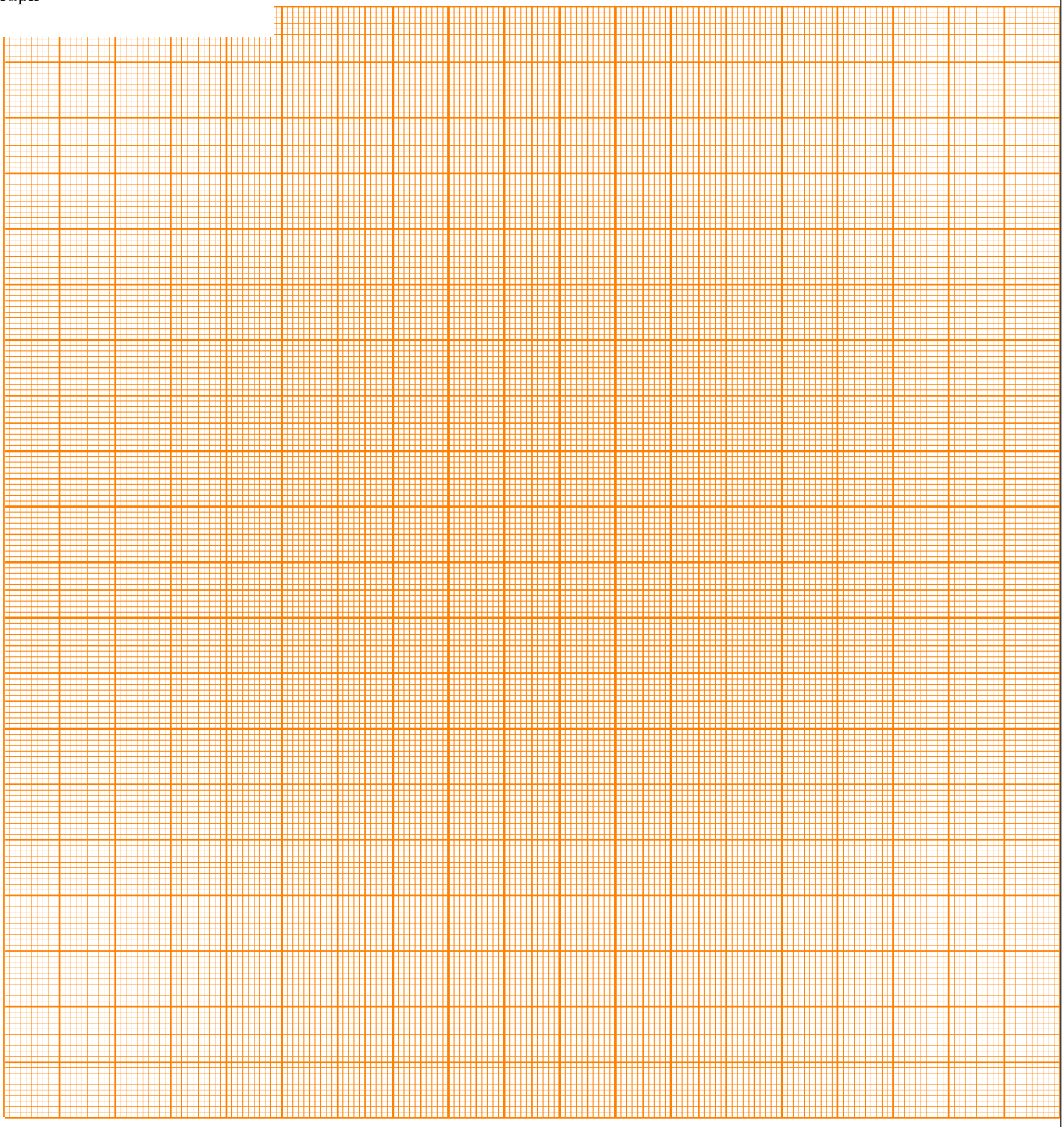
Graph: I versus U

Part D. Emissivity *(3.5 points)*

Formulas used for the graph data:

Calculated data (you don't have to fill the entire table):

Graph



At which temperatures the prediction does hold:

What is the emissivity k in that range:

At which temperatures the prediction does not hold:

Why the prediction fails for these temperatures:

Part E. Specific heat capacity of tungsten *(3.5 points)*

Draw measurement circuit(s):

For quantity of heat Q, directly measured quantities and calculations:

Quantity of heat Q For average specific heat c, measured quantities and calculations:

Average specific heat c

Estimate the magnitudes of the main sources error: